

Below are solutions/hints on some of the questions. There is no guarantee of completeness or correctness; please watch out for typos.

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1. A particle on a one dimensional potential is subject to a potential that is constant for $x \notin [-a, a]$:

$$V(x) = \begin{cases} 0 & \text{for } x < -a \\ \text{unknown} & \text{for } -a < x < a \\ V_0 & \text{for } x > a \end{cases}$$

The Schroedinger equation for this system is found/known to have a solution with energy E with the following forms in the regions outside $[-a, a]$:

$$\begin{aligned} \psi(x) &= \alpha e^{ik_1 x} + \beta e^{-ik_1 x} & \text{for } x < -a \\ \psi(x) &= \gamma e^{ik_2 x} & \text{for } x > a \end{aligned}$$

where

$$\alpha = \frac{k_1 + ik_2}{k_1 - ik_2} \quad \beta = \frac{ik_1}{k_1 - ik_2} \quad \gamma = \frac{\sqrt{k_1 k_2}}{k_1 - ik_2}$$

- (a) [4 pts.] Express k_1 in terms of E .

Solution $\rightarrow$

You can put in the wavefunction in the left ($x < -a$) region into the Schroedinger equation with $V(x) = 0$. This will yield the following relation between E and k_1 . However, you should actually already know the answer, since we worked this out a few times in class.

$$E = \frac{\hbar^2 k_1^2}{2m} \quad k_1 = \sqrt{\frac{2mE}{\hbar^2}}$$

- (b) [4 pts.] Is E larger or smaller than V_0 ? Explain why.

Solution $\rightarrow$

$E > V_0$, because we have a plane wave solution in the region where $V(x) = V_0$. If $E < V_0$, we would have exponential solutions in this region.

- (c) [5 pts.] Express k_2 in terms of E and V_0 .

Solution $\rightarrow$

Again, you can insert $\psi(x) = \gamma e^{ik_2 x}$ in the Schroedinger equation for $x > a$, and this will give the relation

$$E - V_0 = \frac{\hbar^2 k_2^2}{2m} \quad k_2 = \sqrt{\frac{2m}{\hbar^2}(E - V_0)}$$

but you should in fact already know the answer.

- (d) [3 pts.] If we want to interpret this as a scattering problem, which parts of the wavefunction would you interpret as incident wave, reflected wave, and transmitted wave?

Solution $\rightarrow$

incident	$\alpha e^{ik_1 x}$
reflected	$\beta e^{-ik_1 x}$
transmitted	$\gamma e^{ik_2 x}$

- (e) [4 pts.] Calculate the reflection coefficient.

Solution $\rightarrow$

$$R = \left| \frac{\beta}{\alpha} \right|^2 = \left| \frac{ik_1}{k_1 + ik_2} \right|^2 = \frac{k_1^2}{k_1^2 + k_2^2}$$

Note there are no momentum factors. This is explained in the following.

- (f) [5 pts.] Calculate the transmission coefficient.
(Careful: remember momentum factors.)

Solution $\rightarrow$

The momentum factors appear because the transmission coefficient is the ratio of transmitted current to incident current, and the current density carried by a plane wave $Ae^{ipx/\hbar}$ is $|A|^2 p/m$. (Please do prove this to yourself using the definition of the current density.) Thus the transmission coefficient is

$$T = \frac{\text{transmitted current}}{\text{incident current}} = \frac{|\gamma|^2 \hbar k_2 / m}{|\alpha|^2 \hbar k_1 / m} = \left| \frac{\gamma}{\alpha} \right|^2 \times \frac{k_2}{k_1}$$

Note that, in the reflection coefficient, the momentum factors cancel (both are $\hbar k_1$) and hence are not needed.

$$T = \left| \frac{\gamma}{\alpha} \right|^2 \times \frac{k_2}{k_1} = \left| \frac{\sqrt{k_1 k_2}}{k_1 + i k_2} \right|^2 \frac{k_2}{k_1} = \frac{k_2^2}{k_1^2 + k_2^2}$$

- (g) [3 pts.] Check that the sum of the two coefficients makes sense.

Solution $\rightarrow$

$$R + T = \frac{k_1^2}{k_1^2 + k_2^2} + \frac{k_2^2}{k_1^2 + k_2^2} = 1$$

Interpreting the incident plane wave as a steady current/stream of particles, it's obvious that the particles either get reflected or they get transmitted. The sum of the reflected and transmitted currents should be equal to the incident current: otherwise particles are being created or destroyed. This means that the sum of reflection and transmission coefficients needs to be equal to unity.

2. Suppose $\psi_0(x)$ is a properly-normalized wavefunction with

$$\langle \hat{x} \rangle_{\psi_0} = \langle \psi_0 | \hat{x} | \psi_0 \rangle = x_0 \quad \langle \hat{p} \rangle_{\psi_0} = \langle \psi_0 | \hat{p} | \psi_0 \rangle = p_0$$

where x_0 and p_0 are constants. The subscript indicates the state in which the expectation values are being calculated. Define a new wavefunction

$$\psi_{new}(x) = e^{iqx/\hbar} \psi_0(x)$$

where q is a real quantity having dimensions of momentum.

(a) [6 pts.] What is the expectation value $\langle \hat{x} \rangle_{\psi_{new}}$ in the state $\psi_{new}(x)$?

Solution $\rightarrow$

$$\begin{aligned} \langle \hat{x} \rangle_{\psi_{new}} &= \langle \psi_{new} | \hat{x} | \psi_{new} \rangle = \int_{-\infty}^{\infty} x |\psi_{new}(x)|^2 dx \\ &= \int_{-\infty}^{\infty} x |e^{iqx/\hbar} \psi_0(x)|^2 dx = \int_{-\infty}^{\infty} x |e^{iqx/\hbar}|^2 |\psi_0(x)|^2 dx \\ &= \int_{-\infty}^{\infty} x |\psi_0(x)|^2 dx = \langle \hat{x} \rangle_{\psi_0} = x_0 \end{aligned}$$

(b) [7 pts.] What is the expectation value $\langle \hat{p} \rangle_{\psi_{new}}$ in the state $\psi_{new}(x)$?

Solution $\rightarrow$

$$\begin{aligned} \langle \hat{p} \rangle_{\psi_{new}} &= \langle \psi_{new} | \hat{p} | \psi_{new} \rangle = \int_{-\infty}^{\infty} \psi_{new}(x)^* \hat{p} \psi_{new}(x) dx \\ &= -i\hbar \int_{-\infty}^{\infty} \psi_{new}(x)^* \frac{d}{dx} \psi_{new}(x) dx \end{aligned}$$

Now

$$\frac{d}{dx} \psi_{new}(x) = \frac{d}{dx} [e^{iqx/\hbar} \psi_0(x)] = \frac{iq}{\hbar} e^{iqx/\hbar} \psi_0(x) + e^{iqx/\hbar} \frac{d}{dx} \psi_0(x)$$

so that

$$\begin{aligned} \psi_{new}(x)^* \frac{d}{dx} \psi_{new}(x) &= [e^{-iqx/\hbar} \psi_0(x)^*] \left(\frac{iq}{\hbar} e^{iqx/\hbar} \psi_0(x) + e^{iqx/\hbar} \frac{d}{dx} \psi_0(x) \right) \\ &= \frac{iq}{\hbar} |\psi_0(x)|^2 + \psi_0(x)^* \frac{d}{dx} \psi_0(x) \end{aligned}$$

Thus

$$\begin{aligned}\langle \hat{p} \rangle_{\psi_{new}} &= -i\hbar \frac{iq}{\hbar} \int_{-\infty}^{\infty} |\psi_0(x)|^2 dx - i\hbar \int_{-\infty}^{\infty} \psi_0(x)^* \frac{d}{dx} \psi_0(x) dx \\ &= q + \int_{-\infty}^{\infty} \psi_0(x)^* \hat{p} \psi_0(x) dx = q + \langle \hat{p} \rangle_{\psi_0} = q + p_0\end{aligned}$$

- (c) [**2 pts.**] Based on your results, interpret in one sentence the physical significance of multiplying a wavefunction by the factor $e^{iqx/\hbar}$.

Solution $\rightarrow$

Multiplying by this phase factor $e^{iqx/\hbar}$ keeps the average of the position unchanged, and shifts the momentum by the quantity q .

Multiplying by such a phase factor is sometimes informally called ‘boosting’ the wavefunction.

3. Consider a particle with mass m in the state described by wavefunction $\psi(x)$, which we write in amplitude-phase form:

$$\psi(x) = A(x)e^{i\theta(x)}$$

- (a) [**SELF**] Write down the definition of the probability current density.

Solution $\rightarrow$

$$j(x) = \frac{\hbar}{m} \text{Im} \left[\psi(x)^* \frac{\partial}{\partial x} \psi(x) \right]$$

or

$$j(x) = \frac{i\hbar}{2m} \left[\psi(x) \frac{\partial}{\partial x} \psi(x)^* - \psi(x)^* \frac{\partial}{\partial x} \psi(x) \right]$$

The first form is usually easier to calculate with, but of course both should give the same answer. I use the first form below, but students should try out both ways to get the same result.

- (b) [5 pts.] Show that the probability current density is given by

$$j(x) = [A(x)]^2 \frac{\hbar}{m} \partial_x \theta(x)$$

i.e., the probability current is proportional to the *gradient* of the phase.

Solution $\rightarrow$

$$\begin{aligned} \partial_x \psi &= \partial_x [A e^{i\theta}] = (\partial_x A) e^{i\theta} + A \partial_x e^{i\theta} \\ &= (\partial_x A) e^{i\theta} + A e^{i\theta} i \partial_x \theta \end{aligned}$$

Multiplying by $\psi(x)^* = A e^{-i\theta}$,

$$\psi^* \partial_x \psi = A (\partial_x A) + i A^2 \partial_x \theta$$

so that

$$j = \frac{\hbar}{m} \text{Im} [A (\partial_x A) + i A^2 \partial_x \theta] = \frac{\hbar}{m} A^2 \partial_x \theta = \frac{\hbar}{m} [A(x)]^2 \partial_x \theta(x)$$

- (c) [2 pts.] Show that there is no current in a region where the wavefunction is real.

Solution $\rightarrow$

In a region where the function is real, $\theta(x) = 0$, so that $\partial_x \theta(x) = 0$, which implies from the above result:

$$j = 0$$

Alternatively, using the longer definition $j = \frac{i\hbar}{2m} (\psi \partial_x \psi^* - \psi^* \partial_x \psi)$, we find that when the wavefunction is real ($\psi = \psi^*$), the current becomes

$$j = \frac{i\hbar}{2m} (\psi \partial_x \psi - \psi \partial_x \psi) = 0$$